

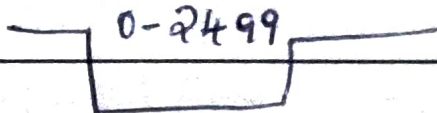
10/07/26

→ UART - Asynchronous → Uses  
different clock signal by transmit  
& receiver.

 Clock.

$$\frac{24M}{9600} = 2500$$

$41.667ns$   
 $\gamma_{clock}^{freq}$

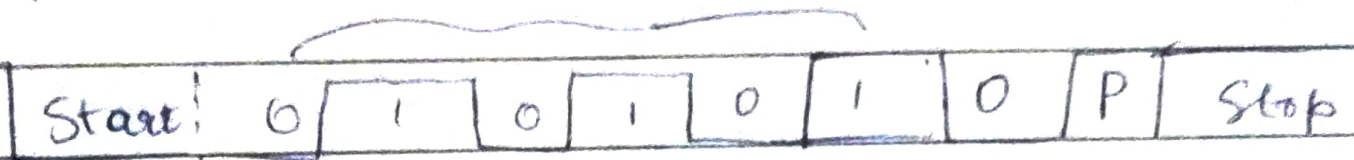
 0-2499

Time period to send 1 bit =  
 $41.667n \times 2500$

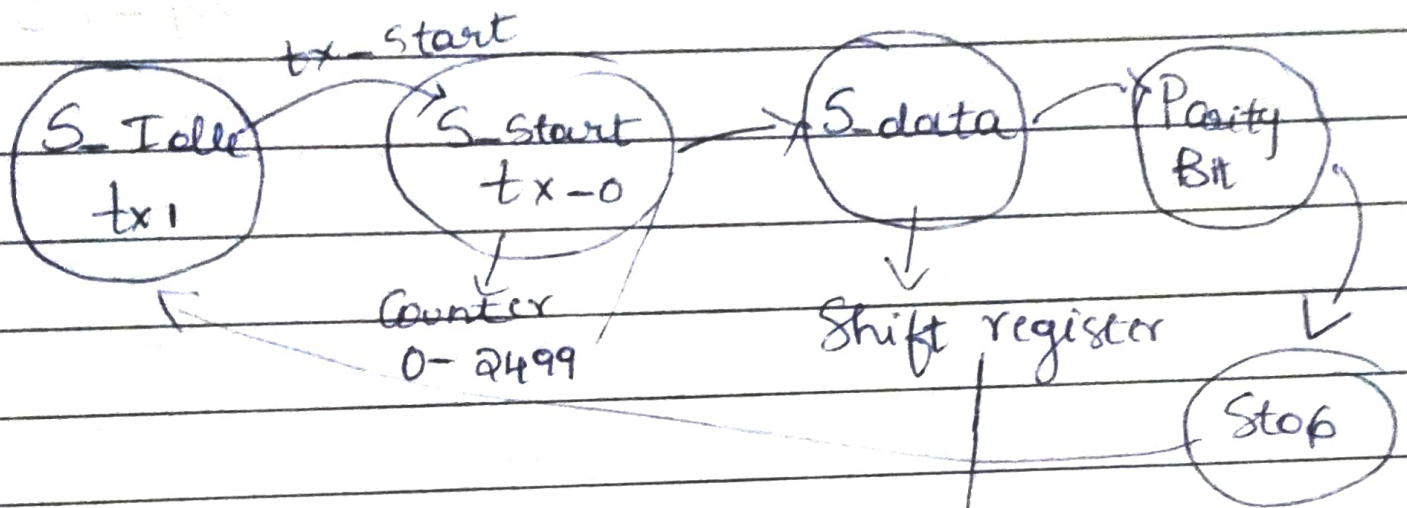
$$= 0.104ms$$

for 9600bits,  $T = 0.10465m \times 9600 = 1.004s$

Parity UART data 7bits



Transmitter Designing Using FSM



$Tx \rightarrow 0 \rightarrow 0-2499$

$Tx \rightarrow 1 \rightarrow 0-2499$

2

$tx = tx\_data[i];$   
 $i = i + 1;$

$\log_2(2499)$

default value of reg is dont care  
"Wire is high impedance"